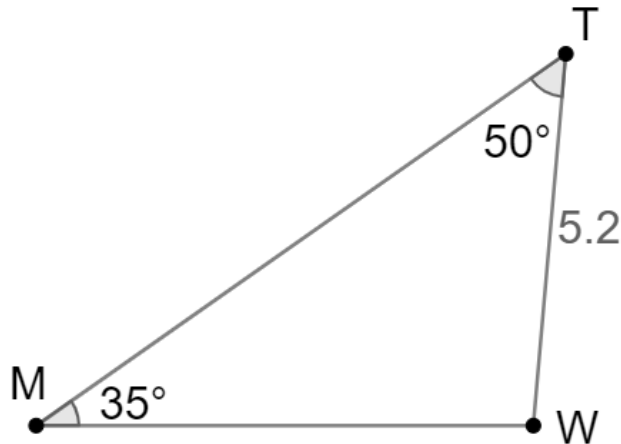


Geometry  
Unit 5  
Lesson 3 Practice

Name \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

Triangle  $TWM$  is shown where  $m\angle M = 35^\circ$ ,  $m\angle T = 50^\circ$ , and  $WT = 5.2$  units.

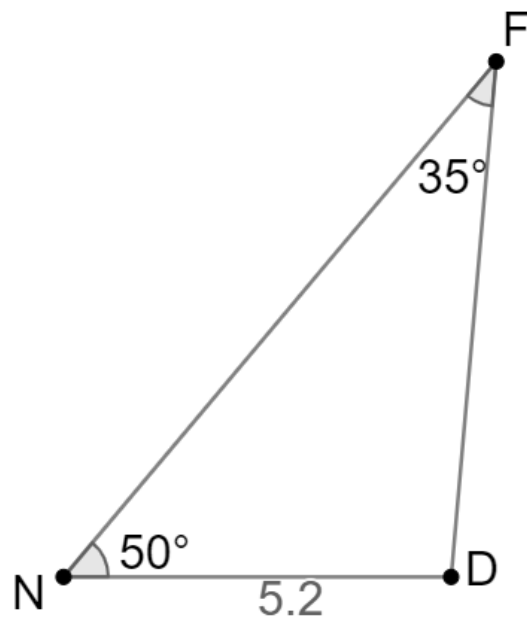


Complete the steps to describe how to construct a triangle that is congruent to  $\triangle TWM$  using the angle-angle-side criteria where the first step is to draw a line that is longer than all of the sides of  $\triangle TWM$ .

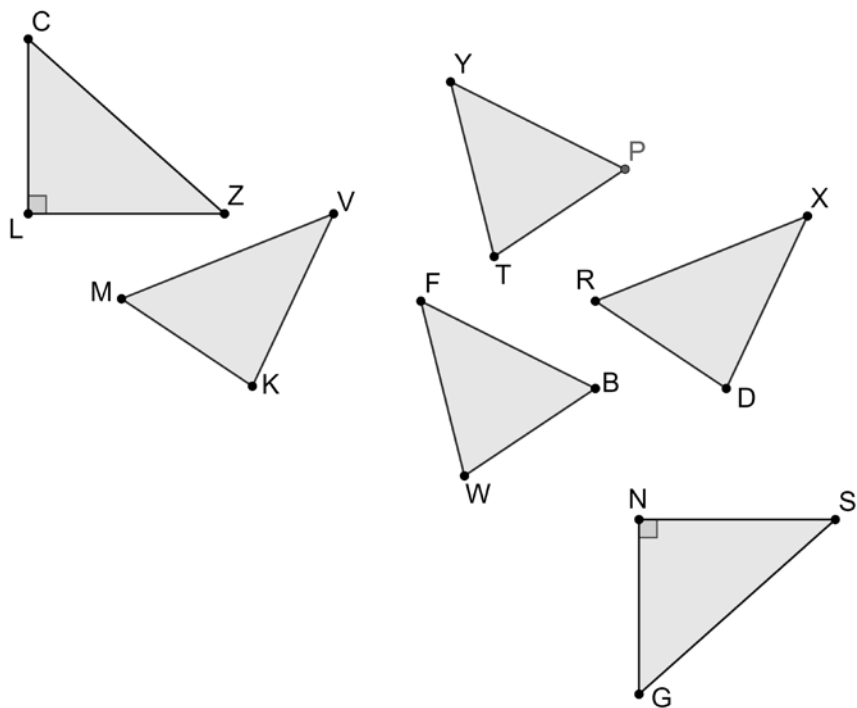
1. Use a compass or straightedge to copy \_\_\_\_\_. Label the endpoints as  $N$  and  $D$ .
2. Use a compass to copy angle \_\_\_\_\_. Construct a congruent angle using point  $N$  as the vertex.
3. Use a compass to copy angle \_\_\_\_\_. Construct a congruent angle so that one side of the angle goes through point  $D$ .

Shanice used the steps to create a triangle that is congruent to  $\triangle TWM$ . Her triangle is shown.

4. Describe what geometric tools Shanice can use to ensure  $\triangle TWM \cong \triangle NDF$ .



Triangles  $CLZ$ ,  $MKV$ ,  $YPT$ ,  $BFW$ ,  $GSN$ , and  $XDR$  are shown.



Determine if the given information is sufficient to determine whether the triangles are congruent. If so, state the congruence theorem or postulate.

|     | Given Information                                                                                               | Are the Triangles Congruent? | Theorem or Postulate Used |
|-----|-----------------------------------------------------------------------------------------------------------------|------------------------------|---------------------------|
| 5.  | $\overline{LZ} \cong \overline{NS}$ , $\overline{CZ} \cong \overline{SG}$ , $\angle LZC \cong \angle NSG$       |                              |                           |
| 6.  | $\angle PTY \cong \angle BWF$ , $\overline{YP} \cong \overline{FB}$ , $\angle TYP \cong \angle WFB$             |                              |                           |
| 7.  | $\overline{VK} \cong \overline{XD}$ , $\overline{MK} \cong \overline{DR}$ , $\angle KVM \cong \angle DXR$       |                              |                           |
| 8.  | $\overline{CZ} \cong \overline{GS}$ , $\overline{CL} \cong \overline{GN}$                                       |                              |                           |
| 9.  | $\overline{MV} \cong \overline{XR}$ , $\overline{KM} \cong \overline{RD}$ , $\overline{XD} \cong \overline{YK}$ |                              |                           |
| 10. | $\angle VMK \cong \angle XRD$ , $\overline{VK} \cong \overline{XD}$ , $\overline{MK} \cong \overline{DR}$       |                              |                           |